Onset-anchored models with delayed transmission

These notes are a companion to the main methodological notes (`notes.tex`) and develop a fully *onset-anchored* branching-process model, in which transmission is referenced to the infector's own symptom onset rather than to their (unobserved) infection time. We build on the main notes throughout: the SSE, SSI and Poisson models of §1, the incubation period and infection-to-onset mapping of §8, the probability of a major outbreak under model uncertainty of §3, and the real-time particle filter of §7. Cross-references below (§and equation numbers) are to the main notes unless stated otherwise.

1 Onset-anchored SSE and SSI models

The bridge model of §8 keeps the generation-time renewal of §1 and adds the incubation period only as an observation delay. Here we go further and anchor transmission directly to the infector's own *symptom onset*. This is appropriate for a disease with negligible pre-symptomatic transmission — Ebola virus disease is our motivating example — for which the onset is the natural reference point for a case's infectiousness.

As in §8 let $D_t \geq 0$ be the number of symptom onsets on day t (the observed quantity) and J_t the number of new infections on day t, with the incubation weights ι_a ($a \geq 1$) mapping infections forward to onsets through the multinomial split (62). In place of the generation interval we introduce the *time from symptom onset to transmission* (TOST) weights τ_s, $s = 0, 1, 2, \ldots$ with $\sum_{s \geq 0} \tau_s = 1$, giving the probability that a transmission event occurs s days after the infector's own onset. Pre-symptomatic transmission would place mass on $s < 0$ and is assumed negligible. The force of infection is now the TOST-weighted sum of past *onset* cohorts. In the onset-anchored SSE model,

$$\lambda_t = \sum_{s=0}^{\infty} \tau_s \, D_{t-s}, \qquad J_t \mid \lambda_t \;\sim\; \mathrm{NB}(\mathrm{mean} = R_0 \lambda_t,\ \mathrm{disp} = k\lambda_t)\,, \tag{1}$$

while in the onset-anchored SSI model each onset cohort carries an aggregate latent infectivity and infections arise as a Poisson process driven by past infectivity,

$$Y_t \mid D_t \;\sim\; \mathrm{Gamma}(\mathrm{shape} = kD_t,\ \mathrm{rate} = k)\,, \qquad \lambda_t = \sum_{s=0}^{\infty} \tau_s \, Y_{t-s}, \qquad J_t \mid \lambda_t \;\sim\; \mathrm{Poisson}(R_0 \lambda_t)\,, \tag{2}$$

exactly as in §1.2 but with the serial-interval weights w_s replaced by the TOST weights τ_s and the driving cohorts taken to be onsets. As $k \to \infty$ both reduce to a common onset-anchored Poisson model with $J_t \sim \mathrm{Poisson}(R_0 \lambda_t)$ and $\lambda_t = \sum_{s \geq 0} \tau_s D_{t-s}$. The generative process alternates $D_t \to \{Y_t\} \to J_t \to D_{>t}$ — with J_t mapped forward to future onsets by (62) — seeded by a single index case with onset on day 0 ($D_0 = 1$).

Indexing convention and causal ordering. Because the incubation weights are supported on $a \geq 1$, an infection on day t cannot produce an onset before day $t+1$. Consequently the onset cohort D_t — and, in the SSI model, its infectivity Y_t — is completely determined by infections on days strictly before t, so it is fully realised before day t's own transmission is evaluated in (1)–(2). This well-orders the recursion. The TOST weights, by contrast, are supported on $s \geq 0$, so a case may transmit on its own onset day. At most one of the two distributions may place mass at lag 0: if both did, a chain of same-day infection, onset and transmission could occur within a single day and the recursion would no longer be well-defined.

Relationship to the serial interval. Consider an infector with onset on day u. It transmits on day $u+S$ with $S \sim \tau$; the infectee, infected on day $u+S$, has onset on day $u+S+A'$ with $A' \sim \iota$. The onset-to-onset serial interval is therefore $S+A'$, distributed as the convolution $\tau * \iota$ with mean $\mathbb{E}[\tau]+\mathbb{E}[\iota]$. Choosing the TOST and incubation means so that they sum to the serial-interval mean used in §1 yields an onset-anchored model that is roughly comparable to the infection-anchored models, while making explicit the intermediate onset-to-transmission and infection-to-onset delays. (This differs from the bridge model of §8, where the onset-to-onset interval is the generation interval convolved with the *difference* of two incubation periods; there the incubation enters only through observation, whereas here the TOST anchors transmission itself.)

2 Probability of a major outbreak

Given an observed onset history $D_0, \ldots, D_r$ (with $D_0 \geq 1$), we again ask for the probability of a major outbreak. Exactly as in the bridge model (§8), and unlike the infection-anchored SSE model of §2.1.1, the observed onsets do not determine the state needed to project the epidemic forward: at the observation horizon there is a pipeline of infected-but-not-yet-symptomatic individuals, whose number and future onset days are latent, together with (in the SSI model) the latent infectivities. We therefore estimate the probability of a major outbreak by the particle filter of §7, matching the observed onset cohorts $D_1, \ldots, D_r$; equivalently, for a single retrospective horizon, by the rejection scheme of §3.2 applied to the onset process. A trajectory is resolved as a major outbreak when a single-day onset count reaches the threshold, and as extinct once no infected individuals remain in the incubation pipeline *and* no onsets have occurred within the TOST support of recent days, so that no further transmission is possible; tracking the pipeline (infections generated minus onsets realised) is necessary because a recent run of zero onsets alone does not preclude onsets already scheduled from earlier infections.

This scheme applies unchanged to the onset-anchored SSE, SSI and Poisson models. Closed-form expressions for the residual extinction probability, analogous to those of §2.1, are likely tractable — the SSE and Poisson offspring still decompose independently, and the SSI infectivities admit the same conditioning arguments — but are not pursued here; the simulation-based estimator suffices for the exploratory comparisons of interest.

Real-time monitoring and model averaging. The bootstrap particle filter of §7 applies directly, advancing onset trajectories one week at a time and retaining the particles whose newly realised onset cohort matches the observation. Because the incubation delay is at least one time unit, the week-t onset cohort is already determined by the state carried in each particle, so conditioning on it requires no fresh simulation. The mean matching fraction at each week is the sequential Monte Carlo estimate of the one-step predictive probability $\mathbb{P}_M(D_w \mid D_{0:w-1})$, and its running product is the marginal likelihood L_M of §3; feeding the per-model evidences through (17)–(18) gives the posterior model probabilities and the model-averaged probability of a major outbreak, tracked in real time as the outbreak unfolds.

Reporting delays (future extension). A natural refinement retains this onset-process model but adds a reporting delay: at the time of analysis some symptom onsets have not yet been reported, so the observed history is a right-truncated and thinned version of the true onset history. The same simulation-based framework accommodates this by matching only the reported subset of onsets (or by marginalising over unreported recent onsets) when accepting trajectories. We note it here as a direction for future work and do not develop it further.